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Assessment task: A3 Part A

Date: 23/05/2026

Portfolio Project – Member 4 (Game Program + Arcade Manager)

Program Title: FunZone Arcade – Number Challenge and Mood Guessing Game

Purpose/Tool:

This activity develops a Python arcade system that combines two fun and interactive games into one program. The first game challenges the user to guess a secret number within limited chances, while the second game tests the user's thinking skills through an mood meaning quiz. The arcade menu allows the user to choose games repeatedly until they decide to exit.

The project demonstrates the use of functions, loops, decision structures, exception handling, lists, random number generation and user interaction in Python.

Tool used: Python 3, Random Library and Jupyter Notebook.

Inputs/Prompt:

Variables:

caleb_secret_number – integer

caleb_attempts – integer

caleb_user_guess – integer

caleb_quiz_list – list

caleb_random_quiz – tuple

caleb_attempts – integer

caleb_user_answer – string

Test values entered:

Menu Choice = 1

Guess = 6

Menu Choice = 2

Answer = sad

Code / AI Output:

In [3]:

```
# ====================================================
# FunZone Arcade Program
# Member 4 – Python Arcade Project
# ====================================================

# Import library
import random

# ----------------------------------------------------
# Function: Number Challenge Game
# ----------------------------------------------------

def caleb_number_game():

    print("===== NUMBER CHALLENGE GAME =====")

    # Generate random number
    caleb_secret_number = random.randint(1, 10)

    # Number of attempts
    caleb_attempts = 3

    # Game loop
    while caleb_attempts > 0:

        try:

            # User input
            caleb_user_guess = int(input("Guess the secret number between 1 and 10:

            # Correct answer
            if caleb_user_guess == caleb_secret_number:
                print("You guessed the correct number.")
                return

            # Hint for user
            elif caleb_user_guess < caleb_secret_number:
                print("Too low. Try again.")

            else:
                print("Too high. Try again.")
```

```
            # Reduce attempts
            caleb_attempts -= 1
            print("Attempts Remaining:", caleb_attempts)

        except ValueError:
            print("Invalid input. Please enter numbers only.")

    print("Game Over.")
    print("The correct number was:", caleb_secret_number)


# ------------------------------------------------------
# Function: Mood Guessing Game
# ------------------------------------------------------

def caleb_mood_game():

    print("\n===== MOOD GUESSING GAME =====")

    # Quiz list
    caleb_quiz_list = [
        ("I feel joyful and smile a lot", "happy"),
        ("I feel relaxed and peaceful", "calm"),
        ("I feel unhappy and low", "sad"),
        ("I have too much energy and excitement", "excited")
    ]

    # Select random quiz
    caleb_random_quiz = random.choice(caleb_quiz_list)

    # Display clue
    print("Guess the mood from this clue:")
    print(caleb_random_quiz[0])

    # Number of attempts
    caleb_attempts = 3

    # Loop until user finishes
    while caleb_attempts > 0:

        # User answer
        caleb_user_answer = input("Enter your answer: ").lower()

        # Correct answer
        if caleb_user_answer == caleb_random_quiz[1]:
            print("Correct answer! Well done.")
            return

        else:
            caleb_attempts -= 1
            print("Incorrect answer.")
            print("Attempts Remaining:", caleb_attempts)

    print("Game Over.")
    print("Correct answer was:", caleb_random_quiz[1])
```

```python
# ----------------------------------------------------
# Arcade Menu Manager
# ----------------------------------------------------

while True:

    print("================================")
    print("WELCOME TO FUNZONE ARCADE")
    print("================================")

    print("1. Number Challenge Game")
    print("2. Mood Guessing Game")
    print("3. Exit")

    try:

        # User menu selection
        caleb_menu_choice = int(input("Enter your choice: "))

        # Run selected game
        if caleb_menu_choice == 1:
            caleb_number_game()

        elif caleb_menu_choice == 2:
            caleb_mood_game()

        elif caleb_menu_choice == 3:
            print("Thank you for playing FunZone Arcade.")
            break

        else:
            print("Invalid choice. Please select from the menu.")

    except ValueError:
        print("Invalid input. Please enter numbers only.")
```

```
================================
WELCOME TO FUNZONE ARCADE
================================
1. Number Challenge Game
2. Mood Guessing Game
3. Exit
===== NUMBER CHALLENGE GAME =====
You guessed the correct number.
================================
WELCOME TO FUNZONE ARCADE
================================
1. Number Challenge Game
2. Mood Guessing Game
3. Exit
===== MOOD GUESSING GAME =====
Guess the mood from this clue:
I feel unhappy and low
```

```
Correct answer! Well done.
================================
WELCOME TO FUNZONE ARCADE
================================
1. Number Challenge Game
2. Mood Guessing Game
3. Exit
Thank you for playing FunZone Arcade.
```

Screenshot (incl. timestamp)

```python
    print("Game Over.")
    print("The correct number was:", caleb_secret_number)


# ----------------------------------------------------
# Function: Mood Guessing Game
# ----------------------------------------------------

def caleb_mood_game():

    print("\n===== MOOD GUESSING GAME =====")

    # Quiz List
    caleb_quiz_list = [
        ("I feel joyful and smile a lot", "happy"),
        ("I feel relaxed and peaceful", "calm"),
        ("I feel unhappy and low", "sad"),
        ("I have too much energy and excitement", "excited")
    ]

    # Select random quiz
    caleb_random_quiz = random.choice(caleb_quiz_list)

    # Display clue
    print("Guess the mood from this clue:")
    print(caleb_random_quiz[0])
```

```python
    # Number of attempts
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        caleb_user_answer = input("Enter your answer: ").lower()

        # Correct answer
        if caleb_user_answer == caleb_random_quiz[1]:
            print("Correct answer! Well done.")
            return

        else:
            caleb_attempts -= 1
            print("Incorrect answer.")
            print("Attempts Remaining:", caleb_attempts)

    print("Game Over.")
    print("Correct answer was:", caleb_random_quiz[1])


# ----------------------------------------------------
# Arcade Menu Manager
# ----------------------------------------------------
```

```python
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# Arcade Menu Manager
# ----------------------------------------------------

while True:

    print("==============================")
    print("WELCOME TO FUNZONE ARCADE")
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    print("1. Number Challenge Game")
    print("2. Mood Guessing Game")
    print("3. Exit")

    try:

        # User menu selection
        caleb_menu_choice = int(input("Enter your choice: "))

        # Run selected game
        if caleb_menu_choice == 1:
            caleb_number_game()

        elif caleb_menu_choice == 2:
            caleb_mood_game()
```

Output Analysis:

The program combines two fun games into one arcade system. The Number Challenge Game allows the user to guess a random number while the Mood Guessing Game asks the user to guess the correct mood from a clue. The program uses loops, functions and decision statements to control the games and check user answers correctly. The output is accurate and the program runs successfully without errors. This activity helped me improve my understanding of Python game programming and user interaction.

In [ ]: